Nafiul Hasan Anik

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RESEARCH INTERESTS

Evaluation and reliability of LLM systems; graph and hypergraph representation learning; grounded retrieval-augmented generation; LLM reasoning about code.

EDUCATION

Bachelor of Science in Computer Science and Engineering *January 2020 – July 2025*
University of Dhaka Dhaka, Bangladesh
CGPA: 3.49/4.00

RESEARCH

Can LLMs Judge Code Without Execution? Predicting Correctness of Submissions from Problem Descriptions and Source Code

Nazira Jesmin Lina, Syed Mumtahin Mahmud, **Nafiul Hasan Anik**, Md Mosaddek Khan, Md. Fahim Arefin
Under review, ACL Rolling Review (October 2026 cycle) • University of Dhaka *May 2026 – October 2026*

- Studied whether LLMs can predict online-judge verdicts from a problem description and source code alone, comparing prompted frontier models with smaller open-weight models fine-tuned with QLoRA.
- Ran the baseline prompting experiments, fine-tuned the smaller models, and analyzed errors by verdict class, finding that models are more reliable on verdicts with visible source-level evidence than on execution-dependent failures.

Overlap-Aware Sub-hypergraph Pattern Frequencies for Hypergraph-Level Classification

Nafiul Hasan Anik, Nazira Jesmin Lina, Md Tanvir Alam, Md. Fahim Arefin
arXiv preprint, 2026 • University of Dhaka *July 2023 – October 2026*

- Extended undergraduate thesis research; proposed overlap-aware frequent sub-hypergraph pattern features for hypergraph-level classification; led feature design, experiments and ablations (relative-frequency vs. presence encodings, classifier and model-selection choices).
- Exceeded the best published mean Macro-F1 on 5 of 6 DHG-Bench datasets (+18.67 on Twitter-Friend, +10.02 on Steam-Player).
- Preprint: arxiv.org/abs/2610.33514 • Code: github.com/NafiHasan/overlap-fhgm

PROFESSIONAL EXPERIENCE

Machine Learning Engineer *April 2025 – Present*
Therap (BD) Ltd. Dhaka, Bangladesh

LLM-as-Judge Evaluation of a Support Chatbot

- Built a pipeline that simulates multi-turn users, who reveal context only when asked, and scores chatbot conversations against curated ground truth with an LLM judge.
- Designed a rubric that scores correctness separately from completeness and accepts any valid branch of multi-path reference solutions.
- Cached opening messages allow paired comparison across 4 chatbot backends; evaluated on 87 curated support questions over 24 tracked runs.

Retrieval-Augmented Support Chatbot

- Built hybrid dense and keyword retrieval over a ∼239K-record knowledge base of user guides, FAQs, issue records and video transcripts, with 7-way LLM query paraphrasing and reranking.
- Built the ingestion pipeline into an Oracle vector store, indexing LLM-generated concept phrases alongside content chunks.
- Added a validator that diagnoses failed answers (invalid question, insufficient context, or response quality) and triggers a matching regeneration strategy.

Faithfulness Evaluation of LLM-Generated Charts

- Built a framework that checks whether LLM-generated charts (Vega-Lite, matplotlib) faithfully represent large nested XML health records.
- Ground truth comes from a deterministic parser cross-validated against an LLM extractor, keeping a value only if both agree or it appears verbatim in the source.
- Recovered the plotted series from chart specifications; LLM-generated code, given the schema but not the values, derives expected values in a sandbox, scored with per-series precision, recall and F1.

RAG Benchmarking and Model Fine-Tuning

- Built a framework to benchmark RAG pipelines from single-passage (SQuAD) to multi-hop (HotpotQA) and book-length (NarrativeQA) question answering.
- Fine-tuned Gemma 3 (1B, 270M) with LoRA, and BERT-family encoders (BERT, RoBERTa, ModernBERT), for classification, question answering and summarization.

ACHIEVEMENTS

- Online judges: Codeforces Expert (max 1719, 1000+ solved); CodeChef 5-star (max rating 2012).
- ACM ICPC Asia Dhaka Regional, 2023: Team placed 19th among 200+ teams.
- SUST Inter-University Programming Contest, 2023: Team placed 15th among 120+ teams.
- Code Samurai, 2024: Team placed 11th in the inter-university hackathon.
- Meta Hacker Cup, 2024: Ranked in the top 22% of Round 2.

TECHNICAL SKILLS

Programming	Python, C++, C, Java
ML / LLMs	PyTorch, TensorFlow, scikit-learn, Hugging Face (Transformers, PEFT), vLLM, LangGraph
Data	NumPy, pandas, Matplotlib
Tools	Git, Docker, FastAPI, LaTeX